Q1) Given

$$\sum_{i=1}^{100}(5a_i + 12) = 4{,}190 \quad and \sum_{j=0}^{98}(a_{j+1} + j) = 5{,}479$$

Find a_{100}

$$\sum_{i=1}^{100}(5a_i + 12) = 4{,}190$$

$$5\sum_{i=1}^{100} a_i + \sum_{i=1}^{100} 12 = 4{,}190$$

$$5\sum_{i=1}^{100} a_i + (100)(12) = 4{,}190$$

$$5\sum_{i=1}^{100} a_i + 1{,}200 = 4{,}190$$

$$\sum_{i=1}^{100} a_i = \frac{4{,}190 - 1{,}200}{4}$$

$$\sum_{i=1}^{100} a_i = 598$$

$$\sum_{j=0}^{98}(a_{j+1} + j) = 5{,}479$$

$$\sum_{j=0}^{98} a_{j+1} + \sum_{j=0}^{98} j = 5{,}479$$

$$\sum_{j=0}^{98} a_{j+1} + 0 + \sum_{j=1}^{98} j = 5{,}479$$

$$\sum_{j=1}^{99} a_j + \frac{(98)(99)}{2} = 5{,}479$$

$$\sum_{j=1}^{99} a_j = 5{,}479 - 4{,}851$$

$$\sum_{j=1}^{99} a_j = 628$$

Combining the two results:

$$a_{100} = \sum_{i=1}^{100} a_i - \sum_{j=1}^{99} a_j$$

$$a_{100} = 598 - 628$$

$$a_{100} = -30$$

Write Final Answer here: $a_{100} = -30$

For **Q2)**, **Q3)** use the following formulas as necessary:

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad\qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

****** *In order to receive marks your approach must rely on sigma notation, summation properties and usage of given formula(s).* ****** *Answers based on combinatorics will not be accepted.*

Q2) Consider the following programming segment:

```
time = 100;
      for (i = 1; i <= n; i++){
            time = time + 2;
            for (k = 1; k <= i+8; k++){
                  for (j = k; j <= 3*k; j++){
                        time = time + 12;
                  }
            }
      }
```

a) Find the value of the variable `time` (in terms of n) after the programming segment is executed. Your formula must be in the simplest form, fully expanded, and terms combined.

(Eg. $time = 7n^2 + 4n + 9$)

$$time = 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}\sum_{j=k}^{3k} 12\right) = 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(3k - k + 1)(12)\right)$$

$$= 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(3k - k + 1)(12)\right) = 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(2k + 1)(12)\right)$$

$$= 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(2k + 1)(12)\right) = 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(24k + 12)\right)$$

$$= 100 + \sum_{i=1}^{n}\left(2 + \sum_{k=1}^{i+8}(24k) + \sum_{k=1}^{i+8}(12)\right) = 100 + \sum_{i=1}^{n}\left(2 + 24\sum_{k=1}^{i+8} k + (i + 8 - 1 + 1)(12)\right)$$

$$= 100 + \sum_{i=1}^{n}\left(2 + 24\frac{(i+8)(i+9)}{2} + (i+8)(12)\right) = 100 + \sum_{i=1}^{n}(2 + 12(i^2 + 17i + 72) + 12i + 96)$$

$$= 100 + \sum_{i=1}^{n}(2 + 12i^2 + 204i + 864 + 12i + 96) = 100 + \sum_{i=1}^{n}(12i^2 + 216i + 962)$$

$$\sum_{i=1}^{n} \qquad (\quad) \qquad -$$

$$= 100 + \sum_{i=1}^{n}(12i^2) + \sum_{i=1}^{n}(216i) + \sum_{i=1}^{n}962 = 100 + 12\sum_{i=1}^{n}i^2 + 216\sum_{i=1}^{n}i + (n - 1 + 1)962$$

$$= 100 + (12)\frac{n(n+1)(2n+1)}{6} + (216)\frac{n(n+1)}{2} + 962n$$

$$= 100 + (2n)(2n^2 + 3n + 1) + (108)(n^2 + n) + 962n$$

$$= 100 + 4n^3 + 6n^2 + 2n + 108n^2 + 108n + 962n$$

$$\therefore time = 4n^3 + 114n^2 + 1072n + 100$$

For `n = 50`

$$time = 4(50)^3 + 114(50)^2 + 1072(50) + 100 = 838{,}700$$

$$\sum_{i=1}^{n} \quad (\ ) \quad -$$

Q3) Consider the following programming segment:

```
time = 500;
      for (i = 1; i <= n; i++){
            time = time + 10;
            for (k = 9; k <= i+3; k++){
                  time = time + 18k;
            }
      }
```

a) Find the value of the variable `time` (in terms of n) after the programming segment is executed. Your formula must be in the simplest form, fully expanded, and terms combined.
(Eg. $time = 7n^2 + 4n + 9$)
Note that the inner loop starts executing for i = 6, adjust the bound as follows

$$time = 500 + \sum_{i=1}^{n} 10 + \sum_{i=6}^{n}\sum_{k=9}^{i+3} 18k = 500 + (n)10 + \sum_{i=6}^{n}\left(18\sum_{k=9}^{i+3} k\right)$$

$$= 500 + 10n + \sum_{i=6}^{n}\left(18\sum_{k=1}^{i+3} k - 18\sum_{k=1}^{8} k\right) = 500 + 10n + \sum_{i=6}^{n}\left((18)\frac{(i+3)(i+4)}{2} - (18)\frac{(8)(9)}{2}\right)$$

$$= 500 + 10n + \sum_{i=6}^{n}(9(i^2 + 7i + 12) - 648) = 500 + 10n + \sum_{i=6}^{n}(9i^2 + 63i - 540)$$

$$= 500 + 10n + \sum_{i=6}^{n} 9i^2 + \sum_{i=6}^{n} 63i - \sum_{i=6}^{n} 540 = 500 + 10n + 9\sum_{i=6}^{n} i^2 + 63\sum_{i=6}^{n} i - (n - 6 + 1)540$$

$$= 500 + 10n + 9\sum_{i=1}^{n} i^2 - 9\sum_{i=1}^{5} i^2 + 63\sum_{i=1}^{n} i - 63\sum_{i=1}^{5} i - (n-5)540$$

$$= 500 + 10n + (9)\frac{n(n+1)(2n+1)}{6} - (9)\frac{5(6)(11)}{6} + (63)\frac{n(n+1)}{2} - (63)\frac{(5)(6)}{2} - 540n + 2{,}700$$

$$= 500 + 10n + \frac{(3n)(2n^2 + 3n + 1)}{2} - 495 + \frac{(63)(n^2 + n)}{2} - 945 - 540n + 2{,}700$$

$$= 10n + \frac{6n^3 + 9n^2 + 3n + 63n^2 + 63n}{2} - 540n + 1{,}760 = 10n + \frac{6n^3 + 72n^2 + 66n}{2} - 540n + 1{,}760$$

$$= 10n + 3n^3 + 36n^2 + 33n - 540n + 1{,}760$$

$$\therefore time = 3n^3 + 36n^2 - 497n + 1{,}760$$

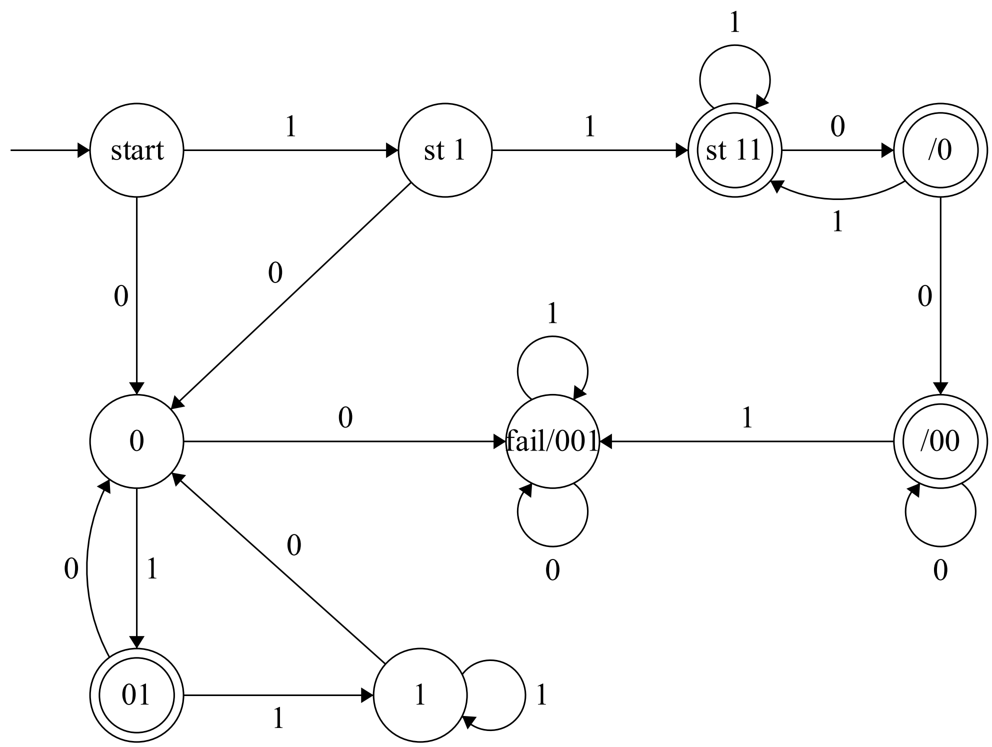
For `n = 50`

$$time = 3(50)^3 + 36(50)^2 - 497(50) + 1{,}760 = 441{,}910$$

$$\sum_{i=1}^{n} \qquad (\;\;) \qquad —$$

- Create FSM drawings at http://madebyevan.com/fsm/, then paste the figure into this document.
- FSM must be clearly labelled and nicely drawn – will not be marked otherwise.
- Make sure there are outgoing 0 and 1 from every state. If that is missing, the machine cannot be marked.
- FSM needs to have the minimum number of required states.
- Properly name the states; S0, S1, S2 or such are not acceptable.
- The start state has to have an arrow pointing to it.
- Efforts must be made to ensure that arcs do not intersect or have their intersections reduced to a minimum. It is easy to move the states around and check.
- Marks will be taken away for missing details.

Q4) Design a finite state machine with input alphabet {0, 1} that accepts all strings that (**start** with 11 or **end** with 01) and do **not contain** 001.



$$\sum_{i=1}^{n} \qquad (\quad) \qquad -$$

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- Marks will be taken away for missing details.

Q5) Design a finite state machine with input alphabet {0, 1} that accepts all strings that (**start** with 01 or **end** with 110) and **contain** an **odd** number of **0's**.